

SARA KIA

AI | Computer Vision | Software Engineering

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GitHub: github.com/sarakia2050 | Portfolio: sarakia2050.github.io | Live Project: ApplyMate Agent (v0-applymate-agent.vercel.app)

PROFESSIONAL SUMMARY

Computer Vision enthusiast with a background in Artificial Intelligence in master and Physics in bachelor, focusing on building real-world image processing and vision-based applications using Python and OpenCV.

Currently strengthening software engineering skills at 42 Heilbronn, with a focus on C programming, Linux, Git, algorithms, and problem solving.

TECHNICAL SKILLS

- **Programming:** Python, C, SQL, basic Java
- **Data & ML:** Pandas, NumPy, Scikit-learn, Statistics, Data Cleaning, Data Visualization, Machine Learning
- **Computer Vision:** OpenCV, Image Processing, Object Detection, OCR basics, Image Preprocessing
- **Software Engineering:** Linux, Git/GitHub, Algorithms, Memory Management, Pointers, Makefile basics
- **AI & Robotics:** SLAM, Evolutionary Algorithms, AI Agents, Robotics Vision
- **Tools:** VS Code, Jupyter Notebook, Google Colab, GitHub, MATLAB
- **Current Focus:** Computer Vision, Machine learning, image processing, AI Agents

SELECTED PROJECTS – AI & COMPUTER VISION

- **Face Detection System (OpenCV, Python):** Developed a real-time face detection system using OpenCV and Haar Cascade classifiers, including image preprocessing and video stream processing.
- **License Plate Detection (Computer Vision):** Developed a computer vision pipeline to detect and extract vehicle license plates using image preprocessing and feature-based techniques.
- **Persian Handwriting Recognition (Computer Vision & ML):** Worked on a computer vision project for recognizing Persian handwritten text using image preprocessing and classification methods.
- **OpenCV Image Processing Pipeline:** Implemented core image processing techniques including grayscale conversion, thresholding, Gaussian blur, edge detection, resizing, and annotation.
- **SpaceX Launch Prediction (Machine Learning):** Built a classification model using structured real-world data, including data cleaning, feature engineering, and evaluation.

PROFESSIONAL EXPERIENCE

Independent Learning & Technical Project Development | 2021 – Present

- Continued developing skills in Python, AI, data analysis, computer vision, and software engineering.
- Built practical portfolio projects and strengthened programming foundations through 42 Heilbronn and self-directed learning.

Programmer – Amiri & Daraei Company | 2013 – 2021

- Worked on software development, technical problem solving, and practical programming tasks.
- Applied programming, analytical thinking, and system design skills in real-world projects.
- Gained experience in Python and data processing, building a strong foundation for later work in machine learning and computer vision.

Physics & Robotics Instructor | 2000–2021

- Taught physics, robotics, and scientific concepts to students.
- Supervised technical and educational projects, including robotics and problem-solving activities.
- Developed strong communication, teaching, analytical thinking, and project guidance skills.

EDUCATION

- **42 Heilbronn – Software Engineering** – 2025 – Present – Project-based software engineering program focused on C programming, Linux, Git, algorithms, problem solving, peer learning, and practical coding projects.
- **M.Sc. Artificial Intelligence** – Amirkabir University of Technology
- **B.Sc. Physics** – University of Tehran

LANGUAGES

Persian – Native | English – Fluent | German – Upper-intermediate

STRENGTHS

Analytical Thinking | Problem Solving | Fast Learning | Structured Working Style | Teamwork | Initiative | Reliability

INTERESTS

Astronomy | Artificial Intelligence | Robotics | Science Communication | Hiking | Yoga | painting